Lecture 6: Sampling Distribution and Normal Distribution

Lecturer

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CLASS CODE

TOAE203

COURSE

Statistics and Probability in Economics and Business

Agenda

01

THE SAMPLING DISTRIBUTION

The room produces the object of the session by hand, before it is named

02

NORMAL DISTRIBUTION & STANDARD ERROR

How far that number moves, and the two quantities that govern the distance

03

WHAT SAMPLE SIZE CANNOT REPAIR

Sample size purchases width. It does not purchase aim

04

THE CENTRAL LIMIT THEOREM

Why the sample mean is bell-shaped when the population is not

05

ASSESSING NORMALITY

The z-score and the Q-Q plot: examining whether a distribution is normal

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THE SAMPLING DISTRIBUTION

The Expected Departure of a Sample Mean

Record an answer before proceeding

(a) under 5% — the sample is a reasonable image of the population

(b) 5–15% — some drift, but the estimate stays usable

(c) 15–30% — the ten firms drawn matter a great deal

(d) over 30% — a single sample says little on its own

Ten firms drawn at random. By how much will your sample mean depart from μ?

Practice 1 — Taking out the sample

Excel · Buoi6-SamplingLab.xlsx · sheet DRAW

Open sheet DRAW. Press F9 exactly once. Read out the large figure when called upon.

TIME

6 minutes

QUANTITY

One draw per participant, n = 10 firms

CONSTRAINT

Each time you press F9, it reset the random seed and discards the sample you are about to report.

Record the value on the board

THE SAMPLING DISTRIBUTION

The Distribution the Room Has Just Produced

$\bar{x}$

the true population mean μ

Not one of the figures on the board is equal to it

$\times\bar{x}$

largest divided by smallest

Two people, the same correct procedure, results differing by that factor

10

estimates, all equally legitimate

None of them is wrong. None of them is the truth

This is a sampling distribution — and it appears in no row and no column of the dataset.

Parameter and Sample Statistic

	Parameter (tham số)	Statistic (thống kê mẫu)
Notation	μ, σ, p (mu, sigma, p)	$\bar{x}$, s, $\hat{p}$
Describes	the **Population**	the **Sample**
In practice	unknown	computed
Today, in this room	known — we declared it	the 10 values on the board
Nature of the object	a fixed number	a random variable

The Sampling Distribution

Definition — The probability distribution of a statistic is called its sampling distribution

1

A sample is only a part of the population, the numerical value of a statistic cannot be expected to give us the exact value of the parameter.

2

The observed value of a statistic depends on the particular sample that happens to be selected.

3

There will be some variability in the value of statistic over different occasions of sampling. Therefore, it is considered to be a random variable

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Practice 2 — Building the Simulation

Excel · sheet SIM10

Use the formula RANDBETWEEN given to take the sample of the firms and calculate the statistics

TIME

6 minutes

QUANTITY

800 rows — each row is one sample of 10 firms

CONSTRAINT

Try out with other default parameters.

Focus on Column A, plus three figures: the mean of the means, their standard deviation, and σ/√n.

The Simulated Result against the Formula

Quantity	Value
Mean of the 800 sample means	___
Population mean μ	___
Standard deviation of those 800 means σ	___
σ/√n – The Standard Error	___

The width of the sampling distribution is settled before any sample is taken.

The Standard Error

DEFINITION

$SE(\bar{x}) = \sigma / \sqrt{n}$

The standard deviation of the sampling distribution. It does not indicate that anything was done incorrectly: it is the unavoidable variation of a correctly obtained estimate.

- σ measures how firms differ from each other
- SE measures how the estimate differs from run to run

WHAT IT DEPENDS ON

Two inputs, and one non-input

Larger σ gives a larger SE. Larger n gives a smaller SE. The size of the population does not enter the formula at all.

- n = 100 from 500 firms, and n = 100 from 100,000 firms give the same standard error

Can you explain why the formula is $\sigma / \sqrt{n}$?

Hint: Mathematics expression

Behaviour under Repeated Sampling

Changes with every new set of samples	Does not change for every samples
Every drawn value	The centre, at μ
The minimum and the maximum	The width, at $\sigma/\sqrt{n}$
The edges of the histogram	The shape

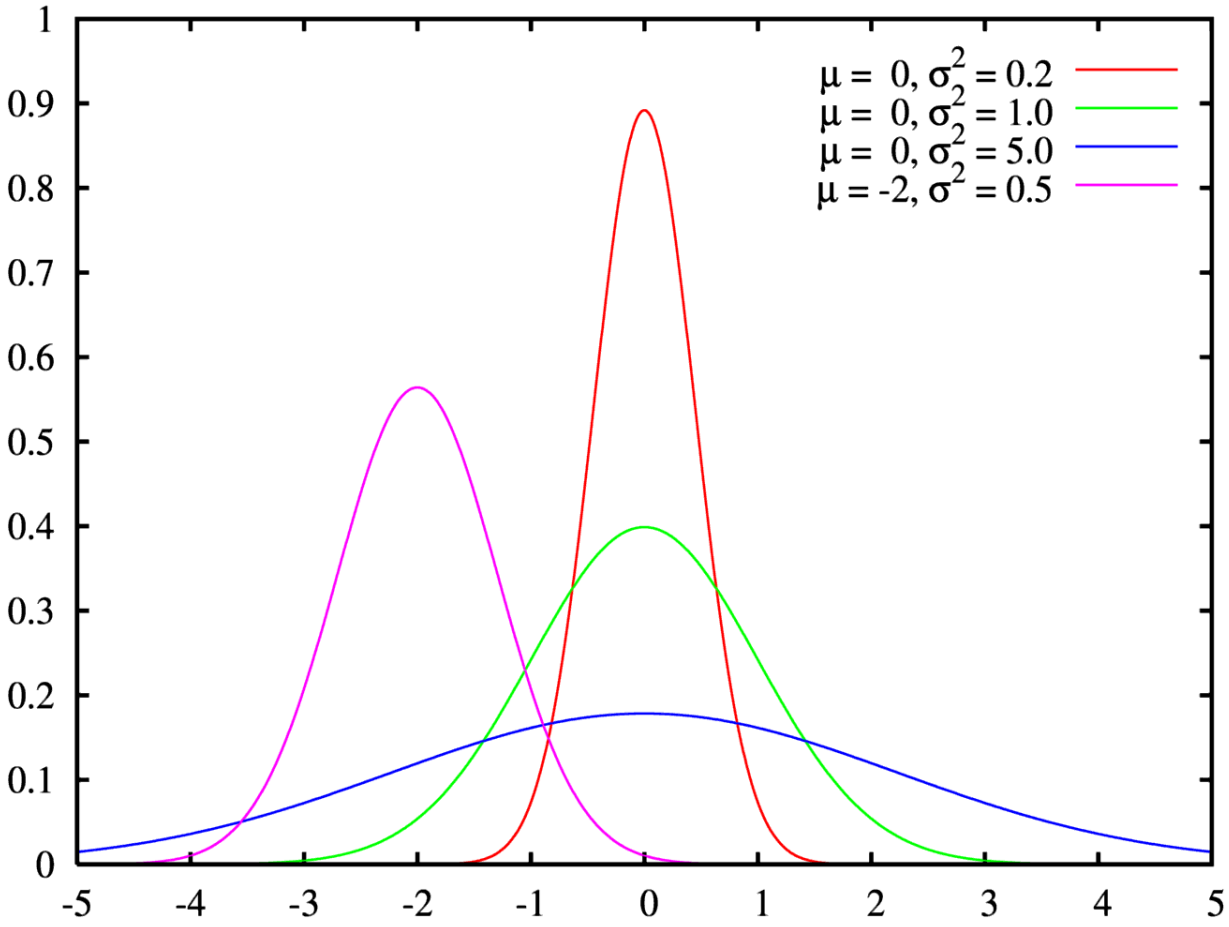
The left side is what one data collection obtains.
The right side is often fixed regardless of the differences in the left side.

The Normal Distribution

The central distinction of this session

A **normal distribution** (also called a **Gaussian distribution** or **Bell curve**) is a continuous probability graph that is symmetric and shaped like a bell.

- Symmetrical shape: The left side mirrors the right side.
- Center peak: The mean (average), median (middle value), and mode (most frequent value) are all equal and sit at the exact center.



Are those all normal distributions? What are the differences?

The Normal Distribution: Two Parameters

MECHANISM

One curve, shifted and rescaled

The normal distribution is determined entirely by 2 parameters: the mean μ, which locates its centre, and the standard deviation σ, which determines its width.

- This is the linear transformation aX + b

CONSEQUENCES

One table of areas suffices

Any statement about any normal distribution reduces to a statement about a single reference curve.

- Two variables in different units become comparable once expressed in units of their own standard deviation
- The sampling distribution built today belongs to this family

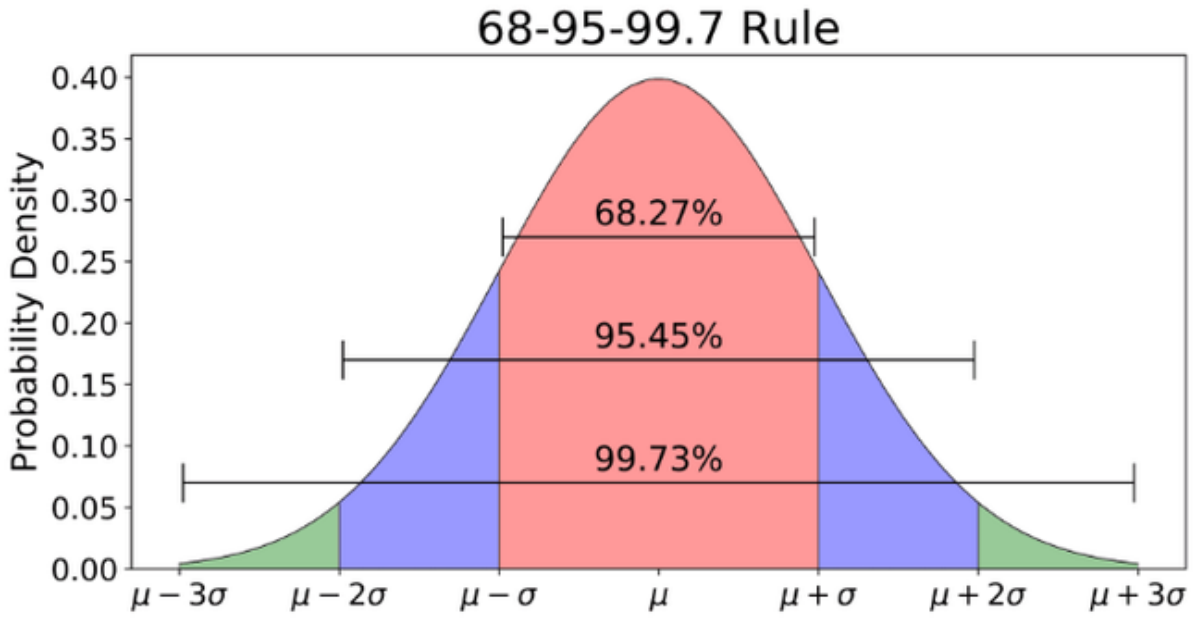
Standardization and the z-score

A COMMON RULER

z = (x − μ) / σ

The **z-score** (điểm z) expresses the distance between a value and the mean in units of the standard deviation. It is the linear transformation **aX + b** with **a = 1/σ**.

- Comparison across variables in different units: a firm at z = 1.5 on ROA and z = 0.2 on revenue growth
- Evaluation of probabilities from one reference distribution, the **standard normal**



The 68-95-99.7 Rule for a Normal Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

The empirical z-score rule

Interval	Proportion of the distribution
$\mu \pm 1\sigma$	approximately 68%
$\mu \pm 2\sigma$	approximately 95%
$\mu \pm 3\sigma$	approximately 99.7%

Z-score Calculator

Interpretation of a z-score: a correction

TRUE WITHOUT ASSUMPTION

A statement about position

The firm lies two standard deviations above the mean. This holds for any distribution whatever, and requires nothing beyond the definitions of μ and σ.

REQUIRES NORMALITY

A statement about area

Under normality the proportion above z = 2 is approximately **2.3%**, not 2%. The resemblance between 2 and 2.3 is a coincidence of digits and not a relationship.

A firm has z = 2 on ROA. Which is correct?

(a) 2% of firms are higher

(b) The firm lies two standard deviations above the mean

(c) Both is correct

The Empirical Rule Applied to the Sampling Distribution

Interval	Predicted	Counted in the 800 means
μ ± 1 SE	68%	___%
μ ± 2 SE	95%	___%
μ ± 3 SE	99.7%	___%

± 1, 2, 3 SE is the width of the answer, and it is available before the study is run.

The Cost of Halving the Standard Error

n = 15 — a modest increase should be enough

n = 20 — double the sample, halve the error

n = 40 — four times the sample

n = 100 — ten times the sample

Remember the formula for Standard Error

The Square-Root Law

To reduce the Standard Error to	Multiply n by
one half	4
one third	9
one tenth	100

Precision is a commodity whose price rises with the square of the precision bought.

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WHAT SAMPLE SIZE CANNOT REPAIR

A Sample Drawn from an Incomplete List

Vote before the reveal

THE DESIGN

Nothing is done incorrectly

A random draw. n = 100. Eight hundred repetitions. Every step of the procedure is performed exactly as before.

- Same procedure, same formula, same sample size

ONE DIFFERENCE

The list contains only profitable firms

The firms that reported a loss were never on the list to begin with. Nothing in the sample reveals that they are missing.

- The case of surveying customers who stayed
- The case of using data on funds still trading

Compared with the unbiased run:

(a) Same center, wider width

(b) Same center, narrower width

(c) Same width, different center

(d) Unchanged

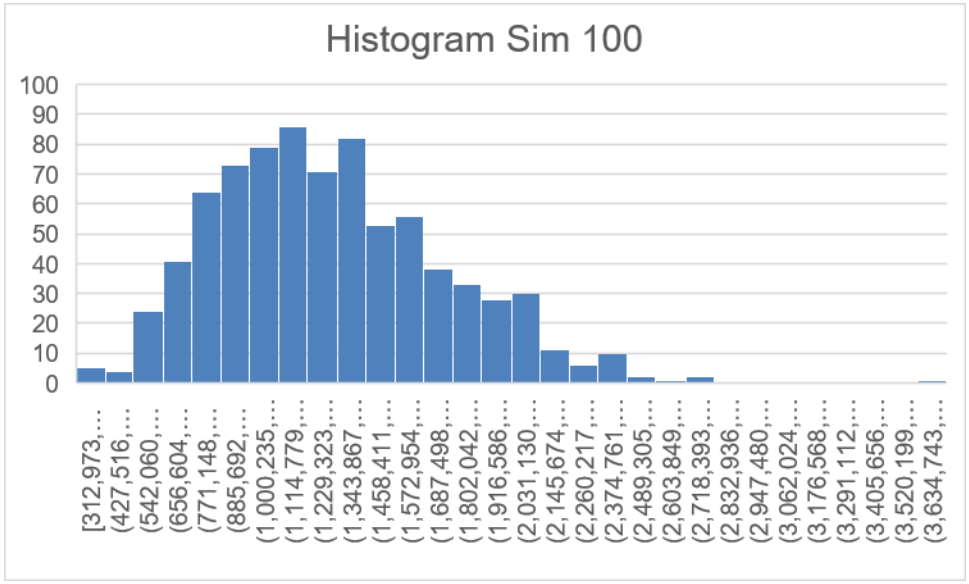
Systematic Departure and Random Variation

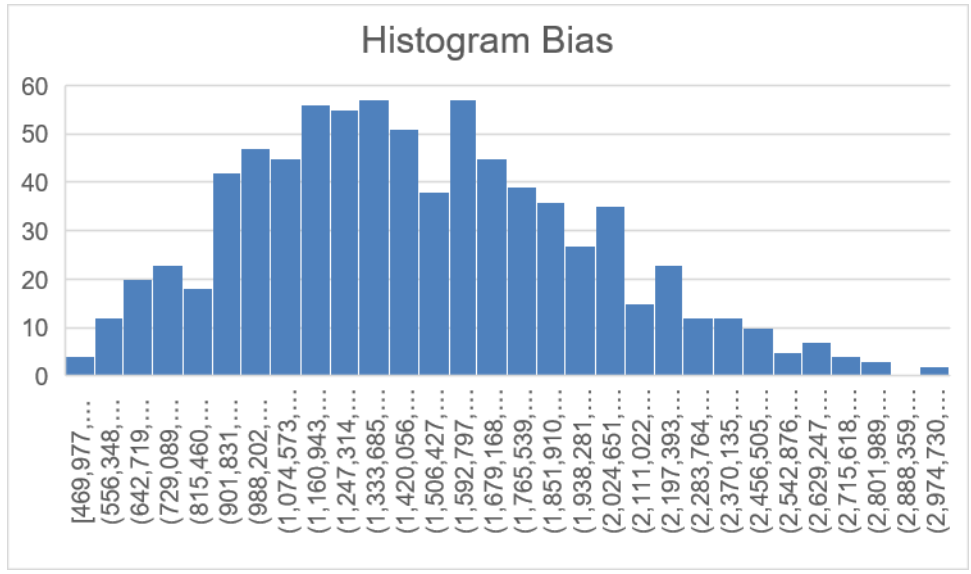
TWO QUANTITIES

Width and centre are separate

Random variation is the width of the sampling distribution, and SE measures it. Bias is the distance from the centre of that distribution to the truth. The second is not a form of the first.

- Complete list: centre ≈ μ, width σ/√n
- Profitable firms only: centre ≈ μ + M , width unchanged
- μ now sits at the edge of the figure, or outside it





WHAT SAMPLE SIZE CANNOT REPAIR

Increasing the Sample Size on a Biased Design

	n = 100	n = 400
Width of the distribution	___	___
Distance of the centre from μ	___	___

Sample size purchases width. It does not purchase aim.

Three Mechanisms of Systematic Departure

Mechanism	Who is excluded	Illustration
Selection bias	units the list never contained	a pay survey run through LinkedIn
Non-response bias	units on the list that did not answer	5,000 sent, 300 returned
Survivorship bias	units that ceased to exist	returns of funds still trading

$\sqrt{n}$ sits in a denominator. Bias sits nowhere in the formula at all.

WHAT SAMPLE SIZE CANNOT REPAIR

Practice 3 — Drawing from an Incomplete List

Excel · a copy of the n = 100 sheet

Replace the source range with the profitable-firms list, leaving every other part of the sheet untouched. Then change n to 400.

TIME

2 minutes

QUANTITY

One changed range, then one changed cell

CONSTRAINT

Both histograms keep the same axis limits, and the true μ is marked on each.

The distance from the centre to μ, at n = 100 and again at n = 400.

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The Shape of a Mean Drawn from a Skewed Population

1

At n = 100, the sample mean will be…

(a) right-skewed as the population is

(b) symmetric and bell-shaped

(c) not predictable

2

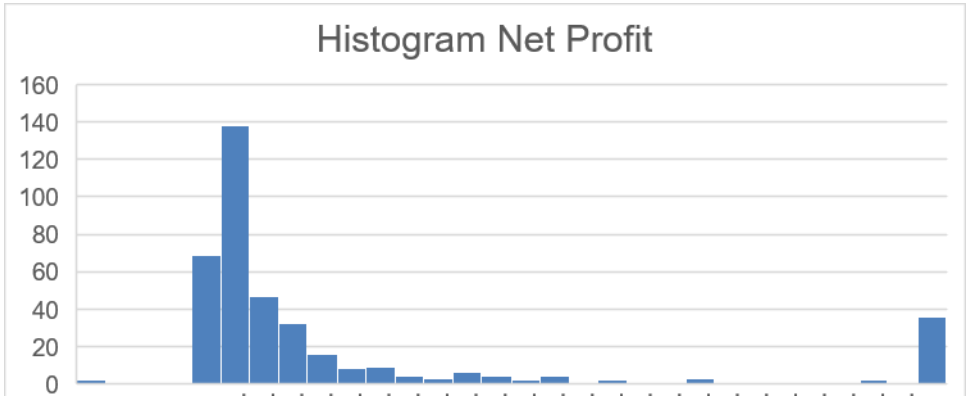
It first looks bell-shaped at…

(a) n = 5

(b) n = 30

(c) n = 100

(d) never, the population being skewed



The population of revenue is strongly right-skewed, with a long upper tail.

Practice 4 — Three Sample Sizes

Excel · sheets SIM5, SIM30, SIM100

Run the machine at n = 5, then 30, then 100. Record the empirical standard error at each, beside σ/√n.

TIME

2 minutes

QUANTITY

Three sample sizes, 800 samples each

CONSTRAINT

One common set of bins and one fixed vertical scale across all three histograms.

Three widths, and the ratio between the first and the last.

The Simulated Result at Three Sample Sizes

n	Empirical SE	σ/√n	Skewness
5	___	___	pronounced right skew
30	___	___	residual skew, visible
100	___	___	approximately symmetric

All three centres sit at μ. The width follows σ/√n. Only the shape changes — and that is the theorem.

What Repeated Sampling Shows

The ordering by width never reverses.

Across repeated sets of samples the three centres do not move, and the widths keep the ratio 1 : 1/√6 : 1/√20.

What is demonstrated is a property of the population and the sample size, not of the particular samples drawn.

What a Single Sample Discloses about Its Own Distribution

IN PRACTICE

One sample is one real study

It gives $\bar{x}$ and s. It does not give σ, it does not give μ, and it does not give the histogram of the sampling distribution.

- Everything else seen today is a luxury of the classroom

THE ESTIMATOR

$s/\sqrt{n}$ in place of $\sigma/\sqrt{n}$

The true standard error needs σ, which is unavailable. The estimated standard error needs the sample alone — and for the row examined, the two agree closely.

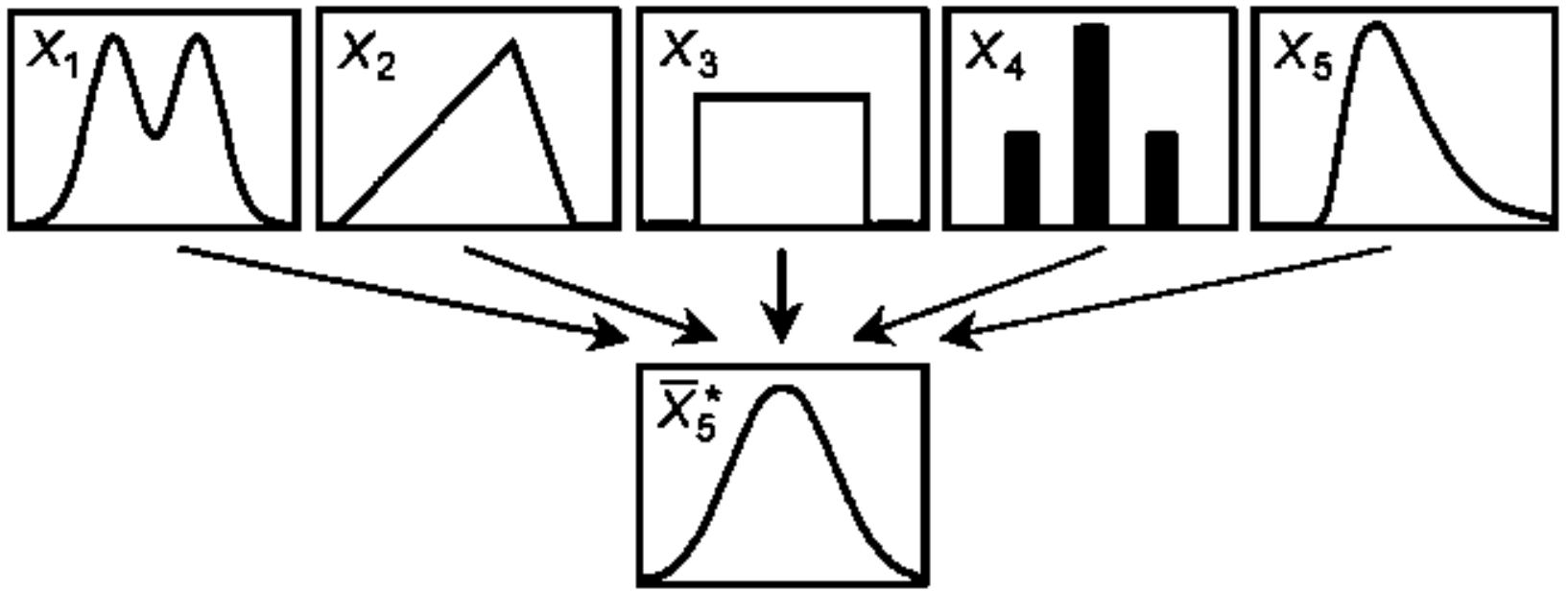
- For one such sample: $s/\sqrt{100}$ = ___ against $\sigma/\sqrt{100}$ = ___

A single sample discloses the width of the distribution it was drawn from. Session 8 begins here.

The Central Limit Theorem

$\bar{x}$ approaches $N(\mu, \sigma^2/n)$ as n grows — whatever the shape of the population.

Required: an independent random sample, and a finite population variance.



The Scope of the Theorem

The theorem states	The theorem does not state
$\bar{x}$ approaches a normal distribution	anything at all, where the sample is not random
Its centre is at μ	that your sample will lies near μ
Its width is $\sigma/\sqrt{n}$	anything about one sample
It holds for every population shape	that n = 30 is always enough

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Comparing Two Distributional Shapes

THE DIFFICULTY

A fitted curve on a histogram decides nothing

The comparison required is between two curved outlines, and visual comparison of curvature is unreliable.

- Similarity of two curves does not appear in that ranking at all
- Human eyes are not as good in identifying curves

THE RESTATEMENT

Are these points collinear?

A Q-Q plot replaces the question do these two shapes agree with the question are these points on a straight line.

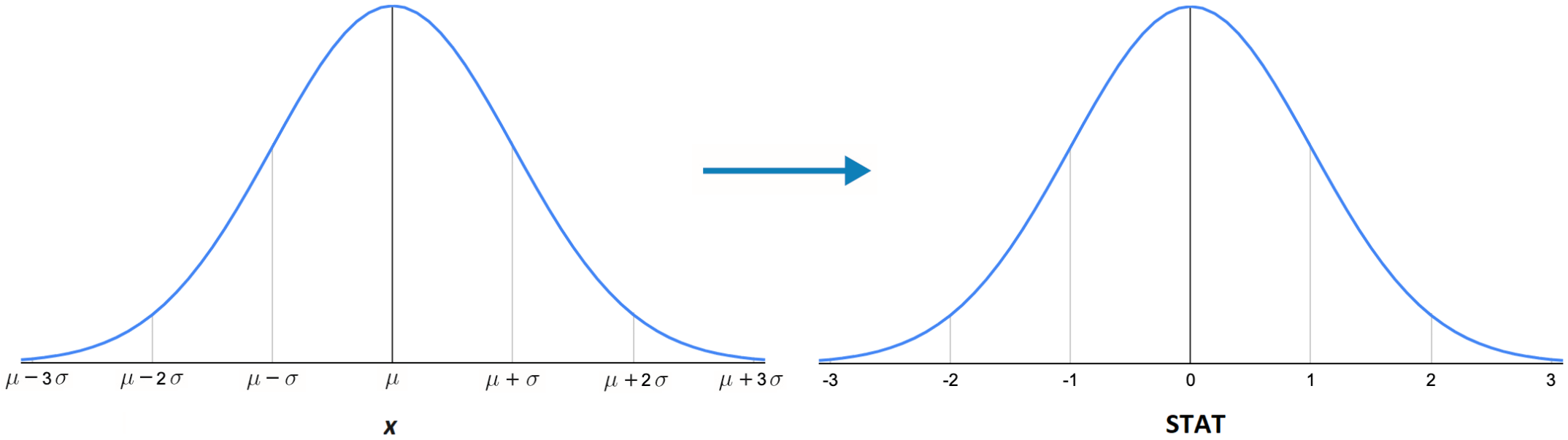
- The second question the human eye answers accurately while the first does not

Standardizing a Sample Mean

$$z = (\bar{x} - \mu) / (\sigma/\sqrt{n})$$

The denominator is the Standard Error, not σ.

The quantity states how many standard errors the estimate lies from the parameter — and under the central limit theorem it is standard normal for large n.



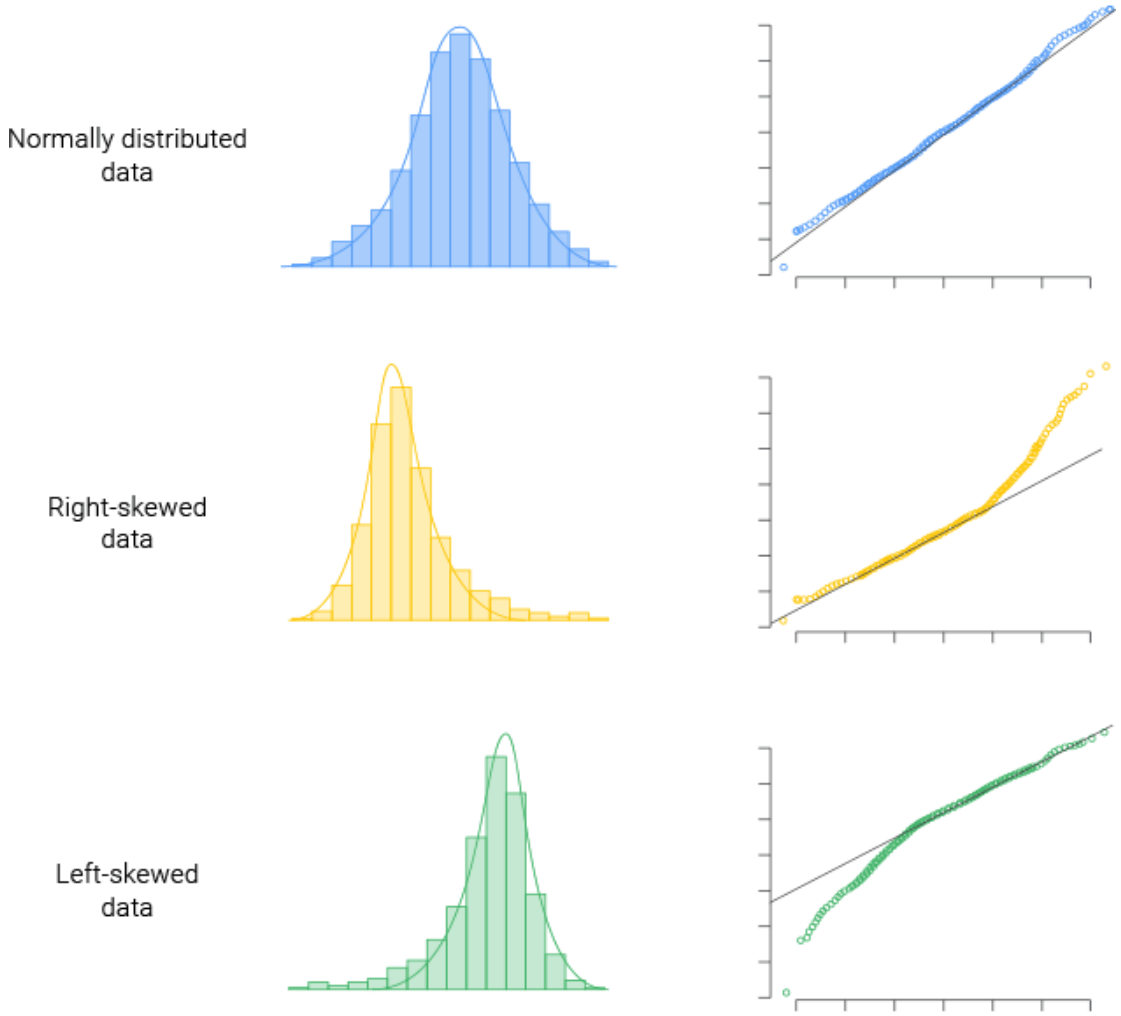
ASSESSING NORMALITY

Q-Q plot

DEFINITION

A **Q-Q plot (quantile-quantile plot)** is a graphical tool used to compare two probability distributions—most commonly a set of sample data against a theoretical normal distribution.

- **X-Axis (Theoretical Quantiles)**: The expected values if your data followed the chosen reference distribution (like a standard normal distribution, N(0,1))
- **Y-Axis (Sample Quantiles)**: Your actual sorted data values.
- **Reference Line**: A diagonal 45-degree line. If the data points fall neatly along this line, the two distributions match



ASSESSING NORMALITY

Practice 5 — A Q-Q Plot by Hand, Then by Machine

Excel · sheet QQ · NORM.S.INV for the theoretical quantile

Build the ten-row table with NORM.S.INV and plot it as a scatter chart. Then repeat for the 800 sample means at each n.

TIME

4 minutes

QUANTITY

One plot of ten points, then three of eight hundred

CONSTRAINT

The horizontal axis carries the theoretical quantile. Plot the points only — add no fitted line.

Four Q-Q plots, and one sentence naming the pattern visible in each.

Construction of a Q-Q Plot

Ten observations, column by column

i	value, ascending	p = (i − 0.5)/n	theoretical quantile z
1	___	0.05	−1.64
2	___	0.15	−1.04
3	___	0.25	−0.67
…	…	…	…
9	___	0.85	+1.04
10	___	0.95	+1.64

The plot is the ten pairs (Value and z). z-score inverts the cumulative normal: from a probability back to a value.

Why a Q-Q Plot Is Straight under Normality

THE ARGUMENT

$$x = \sigma z + \mu$$

If the data are normal, the observed quantile at any position is the standard normal quantile there, transformed by that expression. The transformation is linear, so it plots as a line.

CONSEQUENCES

The line is not fitted

It is the consequence of the assumption under examination. Departure from it says where the assumption fails, not merely whether it fails.

- The ends of the plot carry the tails
- A histogram has too few bars there to support a judgement

The points are collinear because standardisation is linear.

Q-Q Plot Patterns

Pattern	Interpretation
Points collinear	consistent with normality
Right end departs above the line	right-skewed — the largest values exceed what normality allows
Left end departs below the line	left-skewed
Both ends depart outward, an S shape	heavy-tailed — extremes in both directions are too frequent

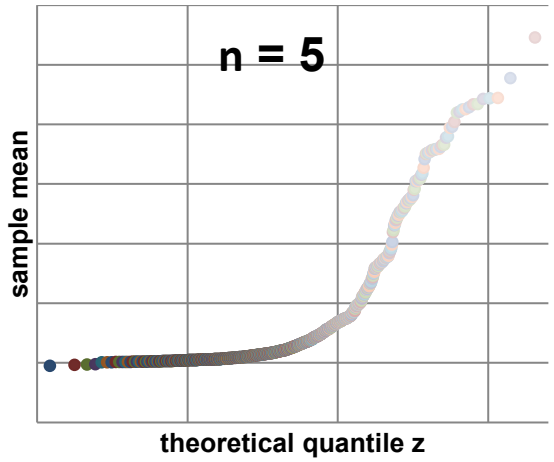
Q-Q Plots of the Three Sampling Distributions

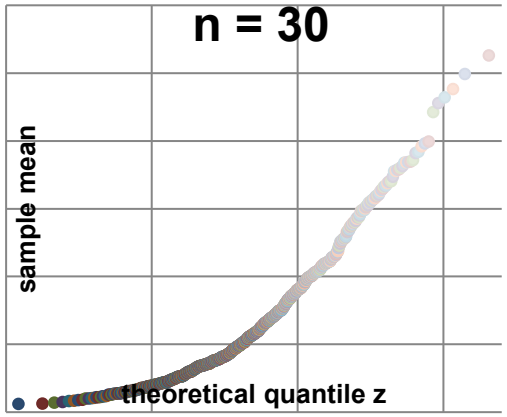
THE SAME DATA, A BETTER ENCODING

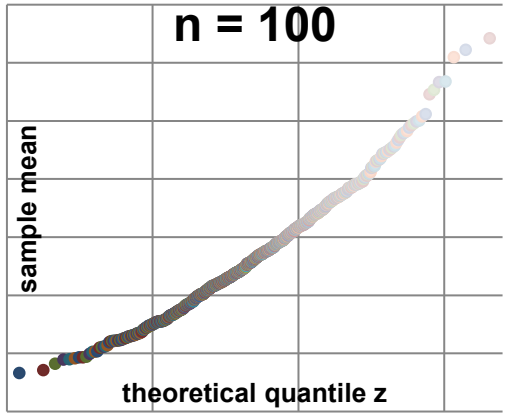
The eight hundred means, replotted

The three columns of sample means built earlier, plotted against the standard normal quantiles instead of drawn as histograms.

- n = 5 — the right end departs markedly above the line
- n = 30 — reduced, and still visible
- n = 100 — approximately collinear







The Sample Size Required for Approximate Normality

Population	n required
already symmetric (ROA)	n = 5 to 10
mildly skewed	n ≈ 30
strongly right-skewed (revenue, assets, income)	n = 100 or more
very heavy-tailed	convergence may be too slow to be of use

The rule is not a number. The rule is: draw the population first.

The Log-Normal Distribution

MECHANISM

Growth is multiplicative

Revenue is bounded below by zero and unbounded above, so a symmetric shape is not available to it. And each year's revenue is the previous year's multiplied by a growth factor.

- Accumulated multiplication is accumulated addition of logarithms
- And by the theorem of minute 96, an accumulated sum tends to a bell

CONSEQUENCES

log(revenue) is the normal one

The upper tail displaces the mean above the bulk of the observations, so the mean is not the typical value. A logarithm also lowers the sample size at which row three of the previous table applies.

- Session 13 builds a regression model on this statement

SUMMARY

Three Things to Take Away

SAMPLING DISTRIBUTION

An estimate is a random variable with a distribution of its own — narrower than the data by √n, and never observed.

SAMPLE SIZE

Precision depends on the number in the sample, not the fraction of the population. Halving the error costs four times the sample.

CENTRAL LIMIT THEOREM

Regardless of the shape of the population, the distribution of $\bar{X}$ is approximately N(μ, $\sigma/\sqrt{n}$), provided that n is large.